Project Initialization and Planning Phase

Date	29 June 2026
Team ID	SWTID-2026-2260
Project	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report :

The proposed project, **Smart Lender – Loan Eligibility Prediction System**, is designed to simplify and improve the loan approval process using machine learning techniques. The system predicts the creditworthiness of loan applicants by analyzing applicant details such as income, education, employment status, loan amount, credit history, and property area. It compares multiple machine learning algorithms including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost to identify the most accurate prediction model. The selected model is integrated into a Flask web application and deployed on IBM Cloud, enabling banks and financial institutions to make faster, more accurate, and data-driven loan approval decisions.

Project Overview	
Objective	To develop an intelligent machine learning-based loan eligibility prediction system that helps banks evaluate loan applications quickly, accurately, and efficiently while reducing manual effort and financial risk.
Scope	The project predicts loan approval using applicant information such as income, education, employment status, loan amount, credit history, and property area. It supports financial institutions by providing real-time loan eligibility predictions through a Flask web application deployed on IBM Cloud.
Problem Statement	

Description	Traditional loan approval processes involve manual verification, making them time-consuming, inconsistent, and prone to human errors. Banks require an automated system that can accurately evaluate loan eligibility using historical data and machine learning techniques.
Impact	Implementing an intelligent loan prediction system improves decision-making, reduces loan processing time, minimizes non-performing assets, enhances customer satisfaction, and increases operational efficiency for financial institutions.
Proposed Solution	
Approach	Develop a machine learning-based web application that preprocesses applicant data, trains multiple classification models, selects the best-performing model (XGBoost), and provides instant loan eligibility predictions through an easy-to-use Flask interface.
Key Features	• Automated loan eligibility prediction using Machine Learning. • Comparison of Decision Tree, Random Forest, KNN, and XGBoost models. • XGBoost selected as the best-performing model. • Real-time prediction through a Flask web application. • User-friendly interface for entering applicant details. • Accurate and fast decision support for banks. • Deployment on IBM Cloud for accessibility and scalability.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i3/i5/i7 Processor (64-bit)
Memory	RAM specifications	4 GB (8 GB Recommended)
Storage	Disk space for data, models, and logs	Minimum 10 GB Free Space (SSD Recommended)
Software		

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost, Pickle
Development Environment	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Data		
Data	Source, size, format	Kaggle dataset